Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

SUMMARY

M.Tech student at IIT Hyderabad with 2+ years of research experience in Artificial Intelligence and Machine Learning, specializing in Computer Vision, OCR-based text and character recognition, and Vision-Language Models. Experienced in image preprocessing, dataset annotation, model benchmarking, and matching extracted text against reference data.

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad Sep 2024 – Jul 2025

Evaluated and fine-tuned Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding and OCR-based text and character recognition, and benchmarked task-specific models (RetinaFace, DeepFace, Siamese networks) for detection, face analysis, and visual tracking.

Junior Research Assistant | TIHAN, IIT Hyderabad: Nov 2023 – Jul 2024

Worked on multimodal perception for autonomous systems by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad: Jul 2023 – Nov 2023

Developed foundational sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

PROJECTS

Vision-Language Model Evaluation for Computer Vision and OCR Tasks (LFOVIA Lab, IIT Hyderabad):

- Evaluated VLMs (Florence-2, Qwen2-VL, Llama 3.2 Vision) for OCR-based text and character recognition, object detection, segmentation, crowd analysis, and action recognition, as alternatives to task-specific models.
- Curated and annotated RGB and depth-mapping datasets, building structured preprocessing, labeling, and quality-validation pipelines that standardized inputs and improved model training consistency and reliability.
- Benchmarked and fine-tuned models using Python-based pipelines, evaluating detection and recognition performance (precision/recall, IoU), and integrated model inference with ROS2 on Linux.

Multi-Channel Conversational AI Platform (IIT Hyderabad Build Program):

- Built a multi-tenant generative-AI platform unifying voice, chat, and email with real-time STT/TTS pipelines.
- Developed a hybrid retrieval pipeline (semantic + keyword) that matches noisy, inconsistently phrased text queries against a reference knowledge base, with LLM-based intent classification, improving answer relevance by 18%.

Camera-Based Perception and Simulation for Autonomous Vehicles (TIHAN, IIT Hyderabad):

- Developed YOLO and DeepSORT perception for real-time object detection and multi-object tracking, tested in CARLA simulation and implemented with ROS2 on Linux; annotated LiDAR and camera data.

THESIS PROJECT

Machine Learning for Non-Oxidative Methane Coupling in a DBD Reactor

- Developed and benchmarked Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict reactor performance from operating conditions with LOOCV validation, reaching cross-validated R^2 up to 0.996.
- Recently adapted the RRSI method to this work to rigorously verify that model improvements are real.

TECHNICAL SKILLS

Programming & Tools: Python, C++, SQL, Linux, ROS2, OpenCV, PyTorch, TensorFlow, CUDA, NumPy, Pandas, Scikit-learn

Computer Vision & OCR: OCR (Text and Character Recognition), Image Preprocessing, Object Detection & Multi-Object Tracking (YOLO, DeepSORT), Face Analysis (DeepFace, RetinaFace), Data Augmentation, Data Annotation (Camera, LiDAR, NIR, Sonar), CNN, CARLA

Generative AI & Multimodal: Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision), Large Language Models (LLMs), LLM Fine-Tuning, RAG, Hybrid Semantic + Keyword Retrieval, Transformer Architectures

Model Evaluation: Precision, Recall, IoU, Model Benchmarking

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs).” IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging.” IEEE i-COSTE 2023, pp. 1–6.

HACKATHONS, ACHIEVEMENTS & LANGUAGES

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad participant

Languages: English, Hindi, Marathi, Japanese (Basic)